MIL-S-21156A(OS)
7 July 1975
SUPERSEDING
MIL-S-21156(WEP)
6 December 1957
and Amendment 2
26 June 1962

MILITARY SPECIFICATION

STEEL BARS, HEAT TREATED (FOR 20-MM AND 30-MM AIRCRAFT GUN BARRELS)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

*1.1 This specification covers heat treated steel bars (2.75 inches nominal dimensions) for the manufacture of 20-mm and 30-mm aircraft gun barrels.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

STANDARDS

Federal

*FED-STD-48	Tolerances for Steel Wrought Products and for Centrifugally Cast Steel
FED-STD-151	Metals; Test Methods

Military

MIL-STD-163	Preparation of Steel Products for Domestic Shipment (Storage) and Overseas Shipment

FSC 9510

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*MIL-STD-1459 Macrograph Standards for Steel Bars, Billets, and Blooms for Ammunition Components

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Dimensions. The dimensions of bars shall be as specified on applicable drawings.

3.2 Material.

3.2.1 Defects. Bars shall be free from cracks. Seams or other surface imperfections shall not exceed the maximum permissible depth stated on the applicable drawing. The contractor shall replace all material found defective after proof firing or on magnetic particle inspection after subsequent machining as specified in the guarantee clause (see 6.1 and 6.2).

*3.2.2 Macrostructure. The macrostructures shall be revealed by macroetch tests (see 4.6.1). The quality and cleanliness of the steel shall be equal to or better than macrographs A4, B3, C8, and with D3 through D8 unacceptable as shown in MIL-STD-1459.

*3.2.3 Chemical composition and hardenability. The contractor shall select any suitable chemical composition provided the hardenability is J55 = 3/16 inch (minimum), to permit induction hardening. Before beginning manufacture of the bars, the contractor shall furnish for review of the contracting officer, a statement of the composition of the steel he proposes to use. This statement shall list the maximum and minimum percentages of carbon, manganese, silicon, the alloying elements intentionally added and the maximum percentages for phosphorus and sulfur. All bars supplied on the contract or order shall have chemical compositions within these approved limits. Government approval of the chemical composition does not relieve the contractor of responsibility for meeting other requirements of this specification.

3.3 Heat treatment. The mechanical properties of the bars shall be obtained by heat treatment. A bar shall be heated uniformly and shall not be in contact with other bars when quenched. The last operation of

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the heat treatment for mechanical properties shall be a tempering treatment which shall consist of uniformly heating the entire bar to a suitable temperature of not less than 1100° Fahrenheit (F).

3.4 Straightening. Bars, after quenching and tempering, shall not have a bow or arc greater than 1/8 inch in any 5 feet. Those having a bow or arc greater than 0.005 inch per inch shall be hot straightened. Those with bows or arcs greater than 0.002 inch per inch and less than 1/4 inch in any 5 feet shall be either cold straightened and stress relieved, or hot straightened.

3.5 Mechanical properties. Tests for evaluation of properties shall be conducted after the final tempering operation. If either cold straightening and stress relieving or hot straightening is required, then tests shall be conducted after these operations.

3.5.1 Yield strength. Yield strength shall be as specified on the applicable drawing.

3.5.2 Reduction of area. The average reduction of area shall comply with the requirements of table I for the average yield strength of the same sample of a heat lot. The interpretation shall be in accordance with 4.6.3.5.

3.5.3 V-notched charpy impact value. The average V-notched charpy impact value shall comply with requirements of table II for the average yield strength of the same batch lot. The interpretation shall be in accordance with 4.6.4.3.

3.5.4 Hardness. Hardness shall be as specified on the applicable drawings.

*3.6 Records. The supplier shall furnish a record of the information needed to comply with this specification. The records of the supplier shall be preserved for one year after the delivery of the last bar of the contract or order.

3.7 Identification marking. Bars shall be marked with lot number (see 4.3) in accordance with MIL-STD-163.

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4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Bar definition. For purposes of inspection, a bar shall be one continuous length which is equal to the ordered length or a multiple thereof.

4.3 Lot.

*4.3.1 Heat lot for macroetch and tension tests, chemical analysis, hardenability, and dimensional and visual inspection. A heat lot shall consist of material from the same melting furnace heat.

4.3.2 Batch lot for impact and hardness tests. A batch lot shall consist of bars from the same melting furnace heat, heat treated at the same time in one furnace, and quenched at the same time in one tank. For continuous processes, batch lots shall consist of bars from the same melting furnace heat which are heat treated in one continuous process. In the latter case, the batch lot shall be limited to the size of the heat unit.

4.4 Sampling.

4.4.1 Macroetch test sample. At least nine billets shall be inspected for compliance of each heat with the macrostructure requirements of 3.2.2. These billets shall be taken from the tops, middles, and bottoms of the first, middle, and last usable ingots cast from each heat lot.

4.4.2 Tension test sample. A maximum of one tension test specimen shall be taken from each bar on first submission for testing. Two bars shall be the minimum number in a sample. Additional tension test specimens may be taken from the same bars on resubmissions. The total number

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of tension test specimens selected per heat lot will depend upon the success attained in satisfying the test requirements of 4.6.3.4 and 4.6.3.5 for each of the sampling groups used under 4.6.3.2.

4.4.3 Impact test sample. A maximum of one impact test specimen shall be taken from each bar on first submission for testing. Two randomly selected bars shall be the minimum number sampled from a batch lot. The total number of impact test specimens selected per batch lot will depend upon the success attained in satisfying the test requirements of 4.6.4.3. If a batch lot is rejected, additional impact test specimens may be taken from the same bars on resubmission. Where there is no corresponding tension test specimen to determine the yield strength of the bar from which an impact test specimen is taken, hardness tests on the impact test specimen may be used to estimate the yield strength by means of table III.

4.4.4 Hardness test sample. A maximum of one test specimen shall be selected from each batch lot for subjection to hardness tests (see 4.6.5).

*4.4.5 Hardenability. Three samples shall be taken from each lot for the hardenability test (see 3.2.3 and 4.6.6).

*4.5 Inspection. Bars shall be inspected dimensionally in accordance with MIL-STD-48 for compliance with 3.1 and 3.4. Bars shall be inspected visually for compliance with the quality provisions of the contract or order and this specification (see 3.2.1). Any lot containing one or more cracked bars shall be subject to 100 percent inspection. Bars shall be inspected for identification markings for conformance to the requirements of 3.7.

4.6 Tests.

*4.6.1 Macroetch tests. The macrostructure of a billet shall be examined by taking a full cross-section which shall be tested in accordance with MIL-STD-1459 to determine compliance with 3.2.2. If one or more billets are rejected as a result of the macroexamination, the portion of the furnace heat lot affected shall be rejected (see 4.4.1). Further testing shall be permitted in order to define better the portion to be rejected.

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*4.6.2 Chemical analysis. The supplier shall take a chemical analysis of each furnace heat of steel in accordance with method 111.1 of FED-STD-151 in order to comply with 3.2.3 and 3.6. The procuring activity shall, at their discretion, call for a check analysis.

4.6.3 Tension tests.

*4.6.3.1 Location of tensile specimen in the bar. Tensile test specimens shall be obtained from the locations indicated on the applicable drawings. If the location is not specified, the test specimens shall be machined from disks obtained at a distance greater than the bar radius from the bar end. The tensile test specimen shall be machined from the disk so that its longitudinal axis is transverse to the bar axis and tangent to the bore periphery of a finished 20-mm gun tube (see figure 1). Type R-1 test specimen shall be prepared in accordance with method 211.1 of FED-STD-151.

4.6.3.2 Location of tensile specimens in the heat. There shall be five independent sampling areas in a furnace heat of steel for tensile properties. At least, two bars shall be tested from each of the five sampling areas which are the top and bottom sixths of the first and last usable ingots and the remaining ingots. In the event of failure of a bar to comply with the requirements of the drawings and this specification, the whole heat shall be rejected. In the event of the inability to identify bars with their position in the heat or in the ingots, tension test sampling shall be conducted on at least five randomly selected bars representing the whole heat.

4.6.3.3 Yield strength determination. Yield strength shall be determined by the offset method as described in FED-STD-151. The offset shall be 0.10 percent (0.001 inch per inch of gage length). The rate of stress application to be followed shall be approved by the procuring activity.

4.6.3.4 Yield strength interpretation. Average yield strength is the arithmetical mean of all yield strength values obtained from the bars of the sampling lot on any submission for testing. All bars shall be considered complying with the requirements if the average yield strength is within the drawing requirements.

4.6.3.5 Reduction of area interpretation. Reduction of area test results shall be evaluated in accordance with table I. If the average of the reduction of area test results is equal to or greater than the

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percentage specified in column A, table I, for the applicable yield strength of the sample lot, the portion of the lot represented shall be acceptable. If the average of test results is equal to or less than the percentage specified in column R, table I, the heat lot represented shall be rejected. Should the average of the test results be between the specified acceptance (column A) and rejection (column R) percentages, additional specimens shall be tested (in increments of one), until a decision can be reached. Each additional test result in conjunction with previous test results shall be used in the computation of a revised sample average for comparison with the appropriate acceptance and rejection of columns of table I.

4.6.4 Impact tests.

4.6.4.1 Making and testing of impact test specimens. Impact specimens shall be V-notched charpy, complying with figure 3, method 221, FED-STD-151. Specimens shall be tested at -65° F.

4.6.4.2 Location of impact test specimens. Charpy V-notched test specimens shall be obtained from locations indicated on the applicable drawings. If no location is specified, the test specimens shall be machined from disks obtained at a distance greater than the bar radius from the end. The specimen shall be machined from the disk so that its longitudinal axis is transverse to the bar axis and the midlength of the notched surface is tangent to the bore periphery of a finished 20-mm gun barrel (see figure 1). If specimens cannot be machined in accordance with 4.6.4.1 because of lack of metal at the corners opposite the notch, test results from such specimens will be acceptable.

4.6.4.3 Charpy V-notch impact test results interpretation. Charpy V-notch impact test results shall be evaluated in accordance with table II. If the average of the charpy V-notch impact test results is equal to or greater than the foot poundage specified in column A of table II for the applicable yield strength and sample size, the lot represented shall be accepted. If this total is equal to or less than the foot poundage specified in column R of table II, the lot represented shall be rejected. Should the average of the test results lie between the specified acceptance (column A) and rejection (column R) foot poundage, additional specimens shall be tested (in increments of one), until a decision can be reached. Each additional test result, in conjunction with previous test results, shall be used in the computation of a revised sample average for comparison with the appropriate acceptance and rejection columns of table II.

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4.6.5 Hardness test. In addition to any locations shown on the applicable drawings, each bar tested for hardness shall be tested at both ends and at the middle to assure the uniformity of the surface hardness of the bar. If all the individual tests of the inspection sample meet all the requirements prescribed by the applicable drawing, all bars in the batch lot shall be considered as having met the requirements for hardness.

*4.6.6 Hardenability. Specimens shall be tested for the hardenability requirement of 3.2.3 in accordance with FED-STD-151, method 711.

4.7 Rejection. Lots shall be rejected, subject to the provisions of 4.8 or 4.9, as applicable, if results of tests specified in 4.6 fail to comply with requirements.

4.8 Defectively machined specimens and faulty testing technique. In the event that defectively machined test specimens or faulty operation of the testing machine or recording device result in the rejection of a batch lot, the test values obtained on such specimens may be discarded, subject to the approval of the procuring activity, and another specimen from the same bar substituted for each such specimen.

4.9 Resubmitting batch or heat lots. Batch or heat lots which have been rejected for either less reduction of area, or impact resistance, or greater or less yield strength or hardness than required, may, at the option of the supplier, be retempered or requenched and retempered, as suitable, then resubmitted for retests in accordance with 4.6, except that macroetch and chemical tests need not be remade. Only the mechanical test results after reheat treatment shall be considered in determining acceptability.

4.10 Test results. Results of all tests made shall be recorded. These records shall include tests taken before and after a retreatment, and any test, the results of which are discarded, shall be recorded with the reasons for discarding the results.

4.11 Acceptance. Lots of bars complying with the requirements of the contract or order, the applicable drawings, specifications, and publications shall be considered acceptable subject to final inspection after proof firing. Bars containing inherent material defects revealed in processing or as a result of proof firing, or material that is not as ordered and specified shall be subject to rejection (see 6.1).

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Steel bars shall be preserved, packaged, and packed in accordance with level A or C as specified in the contract or order (see 6.2).

5.1.1 Level A. Steel bars shall be preserved, packaged, and packed in accordance with MIL-STD-163.

5.1.2 Level C. Steel bars shall be preserved, packaged, and packed according to established commercial practice (see 6.3).

5.2 Marking. In addition to any special markings required by the contract or order shipments shall be marked in accordance with MIL-STD-163.

6. NOTES

6.1 Guarantee and liability. Procurement documents shall specify that any bar which, within 1 year of the date of delivery, is found to contain inherent defects revealed by processing or proof firing or material that is not as ordered and specified shall be rejected and returned to the appropriate supplier for replacement or credit. The contractor shall not be liable for bars which fail on proof firing for reasons other than noncompliance with applicable specifications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Dimensions and tolerances or applicable drawing
c. Guarantee and liability requirements (see 6.1)
d. Maximum permissible depth of defects (see 3.2.1)
e. Level of preservation, packaging, and packing required (see 5.1).

6.3 Commercial preparation for delivery. Commercial practice for preservation, packaging, and packing is described in Department of Commerce Simplified Practice Recommendation 247-52.

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*6.4 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Table I

TRANSVERSE REDUCTION OF AREA VALUES FOR YIELD STRENGTHS INDICATED

Yield strength No. of tests	Transverse reduction of area (%)									
	100,000-104,999 (psi)		105,000-109,999 (psi)		110,000-114,999 (psi)		115,000-119,999 (psi)		120,000-124,999 (psi)	
	A	R	A	R	A	R	A	R	A	R
2	36.79	21.21	34.79	19.21	33.79	18.21	32.79	17.21	31.79	16.21
3	34.20	23.80	32.20	21.80	31.20	20.80	30.20	19.80	29.20	18.80
4	32.90	25.10	30.90	23.10	29.90	22.10	28.90	21.10	27.90	20.10
5	32.12	25.88	30.12	23.88	29.12	22.88	28.12	21.88	27.12	20.88
6	31.60	26.40	29.60	24.40	28.60	23.40	27.60	22.40	26.60	21.40
7	31.23	26.77	29.23	24.77	28.23	23.77	27.23	22.77	26.23	21.77
8	30.95	27.05	28.95	25.05	27.95	24.05	26.95	23.05	25.95	22.05
9	30.73	27.27	28.73	25.27	27.73	24.27	26.73	23.27	25.73	22.27
10	30.56	27.44	28.56	25.44	27.56	24.44	26.56	23.44	25.56	22.44
11	30.42	27.58	28.42	25.58	27.42	24.58	26.42	23.58	25.42	22.58
12	30.30	27.70	28.30	25.70	27.30	24.70	26.30	23.70	25.30	22.70
13	30.20	27.80	28.20	25.80	27.20	24.80	26.20	23.80	25.20	22.80
14	30.11	27.89	28.11	25.89	27.11	24.89	26.11	23.89	25.11	22.89
15	30.04	27.96	28.04	25.96	27.04	24.96	26.04	23.96	25.04	22.96
16	29.97	28.03	27.97	26.03	26.97	25.03	25.97	24.03	24.97	23.03
17	29.92	28.08	27.92	26.08	26.92	25.08	25.92	24.08	24.92	23.08
18	29.87	28.13	27.87	26.13	26.87	25.13	25.87	24.13	24.87	23.13
19	29.82	28.18	27.82	26.18	26.82	25.18	25.82	24.18	24.82	23.18
20	29.78	28.22	27.78	26.22	26.78	25.22	25.78	24.22	24.78	23.22
21	29.74	28.26	27.74	26.26	26.74	25.26	25.74	24.26	24.74	23.26
22	29.71	28.29	27.71	26.29	26.71	25.29	25.71	24.29	24.71	23.29
23	29.68	28.32	27.68	26.32	26.68	25.32	25.68	24.32	24.68	23.32
24	29.65	28.35	27.65	26.35	26.65	25.35	25.65	24.35	24.65	23.35
25	29.00	28.99	27.00	26.99	26.00	25.99	25.00	24.99	24.00	23.99

Note:
A = Average value for acceptance.
R = Average value for rejection.

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Table I (contd)

Yield strength No. of tests	Transverse reduction of area (%)							
	125,000-129,999 (psi)		130,000-134,999 (psi)		135,000-144,999 (psi)		145,000-155,000 (psi)	
	A	R	A	R	A	R	A	R
2	30.79	15.21	28.79	13.21	27.79	12.21	26.79	11.21
3	28.20	17.80	26.20	15.80	25.20	14.80	24.20	13.80
4	26.90	19.10	24.90	17.10	23.90	16.10	22.90	15.10
5	26.12	19.88	24.12	17.88	23.12	16.88	22.12	15.88
6	25.60	20.40	23.60	18.40	22.60	17.40	21.60	16.40
7	25.23	20.77	23.23	18.77	22.23	17.77	21.23	16.77
8	24.95	21.05	22.95	19.05	21.95	18.05	20.95	17.05
9	24.73	21.27	22.73	19.27	21.73	18.27	20.73	17.27
10	24.56	21.44	22.56	19.44	21.56	18.44	20.56	17.44
11	24.42	21.58	22.42	19.58	21.42	18.58	20.42	17.58
12	24.30	21.70	22.30	19.70	21.30	18.70	20.30	17.70
13	24.20	21.80	22.20	19.80	21.20	18.80	20.20	17.80
14	24.11	21.89	22.11	19.89	21.11	18.89	20.11	17.89
15	24.04	21.96	22.04	19.96	21.04	18.96	20.04	17.96
16	23.97	22.03	21.97	20.03	20.97	19.03	19.97	18.03
17	23.92	22.08	21.92	20.08	20.92	19.08	19.92	18.08
18	23.87	22.13	21.87	20.13	20.87	19.13	19.87	18.13
19	23.82	22.18	21.82	20.18	20.82	19.18	19.82	18.18
20	23.78	22.22	21.78	20.22	20.78	19.22	19.78	18.22
21	23.74	22.26	21.74	20.26	20.74	19.26	19.74	18.26
22	23.71	22.29	21.71	20.29	20.71	19.29	19.71	18.29
23	23.68	22.32	21.68	20.32	20.68	19.32	19.68	18.32
24	23.65	22.35	21.65	20.35	20.65	19.35	19.65	18.35
25	23.00	22.99	21.00	20.99	20.00	19.99	19.00	18.99

Note:
A = Average value for acceptance.
R = Average value for rejection.

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Table II

TRANSVERSE V-NOTCHED CHARPY IMPACT VALUES FOR YIELD STRENGTHS INDICATED

Yield strength No. of tests	Transverse V-notched charpy impact (ft-lb)									
	100,000-104,999 (psi)		105,000-109,999 (psi)		110,000-114,999 (psi)		115,000-119,999 (psi)		120,000-124,999 (psi)	
	A	R	A	R	A	R	A	R	A	R
2	33.12	21.68	31.12	19.68	29.12	17.68	27.17	16.43	25.17	14.43
3	31.21	23.59	29.21	21.59	27.21	19.59	25.38	18.22	23.38	16.22
4	30.26	24.54	28.26	22.54	26.26	20.54	24.48	19.12	22.48	17.12
5	29.69	25.11	27.69	23.11	25.69	21.11	23.95	19.65	21.95	17.65
6	29.30	25.50	27.30	23.50	25.30	21.50	23.59	20.01	21.59	18.01
7	29.03	25.77	27.03	23.77	25.03	21.77	23.33	20.27	21.33	18.27
8	28.83	25.97	26.83	23.97	24.83	21.97	23.14	20.46	21.14	18.46
9	28.67	26.13	26.67	24.13	24.67	22.13	22.99	20.61	20.99	18.61
10	28.54	26.26	26.54	24.26	24.54	22.26	22.87	20.73	20.87	18.73
11	28.44	26.36	26.44	24.36	24.44	22.36	22.78	20.82	20.78	18.82
12	28.35	26.45	26.35	24.45	24.35	22.45	22.69	20.91	20.69	18.91
13	28.28	26.52	26.28	24.52	24.28	22.52	22.63	20.97	20.63	18.97
14	28.22	26.58	26.22	24.58	24.22	22.58	22.57	21.03	20.57	19.03
15	28.16	26.64	26.16	24.64	24.16	22.64	22.52	21.08	20.52	19.08
16	28.11	26.69	26.11	24.69	24.11	22.69	22.47	21.13	20.47	19.13
17	28.07	26.73	26.07	24.73	24.07	22.73	22.43	21.17	20.43	19.17
18	28.04	26.76	26.04	24.76	24.04	22.76	22.40	21.20	20.40	19.20
19	28.00	26.80	26.00	24.80	24.00	22.80	22.36	21.24	20.36	19.24
20	27.97	26.83	25.97	24.83	23.97	22.83	22.34	21.26	20.34	19.26
21	27.94	26.86	25.94	24.86	23.94	22.86	22.31	21.29	20.31	19.29
22	27.92	26.88	25.92	24.88	23.92	22.88	22.29	21.31	20.29	19.31
23	27.90	26.90	25.90	24.90	23.90	22.90	22.27	21.33	20.27	19.33
24	27.88	26.92	25.88	24.92	23.88	22.92	22.25	21.35	20.25	19.35
25	27.40	27.39	25.40	25.39	23.40	23.39	21.80	21.79	19.80	19.79

Note:
A = Average value for acceptance.
R = Average value for rejection.

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Table II (contd)

Yield strength / No. of tests	Transverse V-notched charpy impact (ft-lb)									
	125,000-129,999 (psi)		130,000-134,999 (psi)		135,000-139,999 (psi)		140,000-144,999 (psi)		145,000-150,000 (psi)	
	A	R	A	R	A	R	A	R	A	R
2	24.22	14.18	22.22	12.18	20.28	10.92	18.28	8.92	17.33	8.67
3	22.55	15.85	20.55	13.85	18.72	12.48	16.72	10.48	15.89	10.11
4	21.71	16.69	19.71	14.69	17.94	13.26	15.94	11.26	15.16	10.84
5	21.21	17.19	19.21	15.19	17.47	13.73	15.47	11.73	14.73	11.27
6	20.87	17.53	18.87	15.53	17.16	14.04	15.16	12.04	14.44	11.56
7	20.63	17.77	18.63	15.77	16.94	14.26	14.94	12.26	14.24	11.76
8	20.46	17.94	18.46	15.94	16.77	14.43	14.77	12.43	14.08	11.92
9	20.32	18.08	18.32	16.08	16.64	14.56	14.64	12.56	13.96	12.04
10	20.20	18.20	18.20	16.20	16.53	14.67	14.53	12.67	13.87	12.13
11	20.11	18.29	18.11	16.29	16.45	14.75	14.45	12.75	13.79	12.21
12	20.04	18.36	18.04	16.36	16.38	14.82	14.38	12.82	13.72	12.28
13	19.97	18.43	17.97	16.43	16.32	14.88	14.32	12.88	13.67	12.33
14	19.92	18.48	17.92	16.48	16.27	14.93	14.27	12.93	13.62	12.38
15	19.87	18.53	17.87	16.53	16.22	14.98	14.22	12.98	13.58	12.42
16	19.83	18.57	17.83	16.57	16.18	15.02	14.18	13.02	13.54	12.46
17	19.79	18.61	17.79	16.61	16.15	15.05	14.15	13.05	13.51	12.49
18	19.76	18.64	17.76	16.64	16.12	15.08	14.12	13.08	13.48	12.52
19	19.73	18.67	17.73	16.67	16.09	15.11	14.09	13.11	13.46	12.54
20	19.70	18.70	17.70	16.70	16.07	15.13	14.07	13.13	13.43	12.57
21	19.68	18.72	17.68	16.72	16.04	15.16	14.04	13.16	13.41	12.59
22	19.66	18.74	17.66	16.74	16.02	15.18	14.02	13.18	13.39	12.61
23	19.64	18.76	17.64	16.76	16.01	15.19	14.01	13.19	13.38	12.62
24	19.62	18.78	17.62	16.78	15.99	15.21	13.99	13.21	13.36	12.64
25	19.20	19.19	17.20	17.19	15.60	15.59	13.60	13.59	13.00	12.99

Note:

A = Average value for acceptance.

R = Average value for rejection.

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Table III

MINIMUM HARDNESS REQUIREMENTS AT COMPARABLE STRENGTH LEVELS

Tensile strength (psi)	Yield strength 0.10% offset (psi)	Brinell hardness number	Brinell indentation diameter (mm)	Rockwell hardness number (C scale)
121,000 - 125,999	100,000 - 104,999	247 - 254	3.85 - 3.81	24 - 25
126,000 - 130,499	105,000 - 109,999	255 - 264	3.80 - 3.73	25 - 27
130,500 - 134,999	110,000 - 114,999	265 - 272	3.72 - 2.68	27 - 28
135,000 - 139,999	115,000 - 119,999	273 - 282	3.67 - 3.62	28 - 29.5
140,000 - 144,999	120,000 - 124,999	283 - 292	3.61 - 3.56	29.5 - 31
145,000 - 149,999	125,000 - 129,999	293 - 301	3.55 - 3.51	31 - 32
150,000 - 154,999	130,000 - 134,999	302 - 310	3.50 - 3.46	32 - 33
155,000 - 159,999	135,000 - 139,999	311 - 320	3.45 - 3.41	33 - 34.5
160,000 - 164,999	140,000 - 144,999	321 - 329	3.40 - 3.37	34.5 - 35.5
165,000 - 169,999	145,000 - 149,999	330 - 340	3.36 - 3.31	35.5 - 36.6

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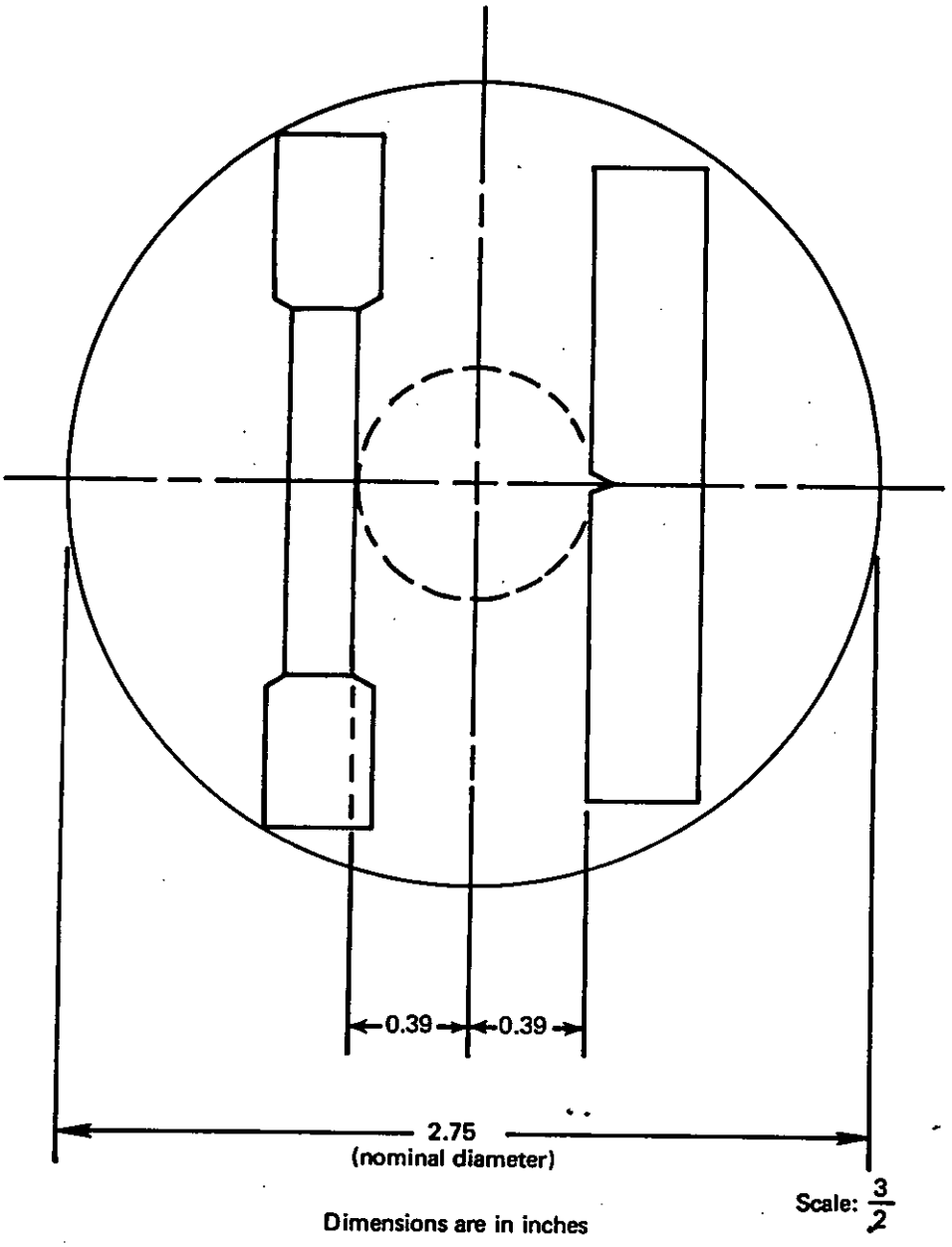


FIGURE 1. LOCATION OF TRANSVERSE MECHANICAL SPECIMENS IN TEST DISK

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DOCUMENT IDENTIFIER AND TITLE: MIL-S-21156A(OS), Steel Bars, Heat Treated (For 20-mm and 30-mm Aircraft Gun Barrels)

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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